

Submission STAT1-PS2-SUB06

Question STAT1-PS2-Q01

notation changes halfway:

1. $\bar{x} \approx 33/5 \approx 6.60$
2. $s^2 \approx \Sigma(x_i - \bar{x})^2 / (5-1) \approx 16.30$

marks, subscripts, and units are mixed up

result written as: mean=6.60; sample variance=16.30 (please interpret)

Question STAT1-PS2-Q02

notation changes halfway:

1. $P(D_1 \cap D_2) \approx (5/13)((5-1)/(13-1))$
2. ≈ 0.1282 ; the population shrinks after draw 1

marks, subscripts, and units are mixed up

result written as: 0.1282 (please interpret)

Question STAT1-PS2-Q03

notation changes halfway:

1. $SE \approx \sqrt{(p(1-p)/n)} \approx \sqrt{(0.55(1-0.55)/40)} \approx 0.0787$
2. $CI \approx 0.55 \pm 1.96(0.0787) \approx (0.396, 0.704)$
3. In repeated samples, the procedure captures the population proportion about 95% of the time.

marks, subscripts, and units are mixed up

result written as: (0.396, 0.704); repeated-sampling interpretation (please interpret)

Question STAT1-PS2-Q04

notation changes halfway:

1. Increase sample size n to reduce standard error.
2. Reduce confidence level or population variability when justified.
3. Lower confidence changes coverage, so precision is not free.

marks, subscripts, and units are mixed up

result written as: increase n ; lower confidence level or variability; note the coverage trade-off (please interpret)

Question STAT1-PS2-Q05

notation changes halfway:

1. $0.03 < 0.05$, so reject H_0 under the stated rule.
2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

marks, subscripts, and units are mixed up

result written as: reject H_0 ; evidence against H_0 , not proof that H_0 is false (please interpret)

Question STAT1-PS2-Q06

notation changes halfway:

1. $\bar{x} \approx 29/5 \approx 5.80$
2. $s^2 \approx \Sigma(x_i - \bar{x})^2 / (5-1) \approx 16.70$

marks, subscripts, and units are mixed up

result written as: mean=5.80; sample variance=16.70 (please interpret)

Question STAT1-PS2-Q07

notation changes halfway:

1. $P(D_1 \cap D_2) \approx (5/19)((5-1)/(19-1))$

2. ≈ 0.0585 ; the population shrinks after draw 1

marks, subscripts, and units are mixed up

result written as: 0.0585 (please interpret)

Question STAT1-PS2-Q08

notation changes halfway:

1. $SE \approx \sqrt{p(1-p)/n} \approx \sqrt{(0.45(1-0.45)/80)} \approx 0.0556$

2. $CI \approx 0.45 \pm 1.96(0.0556) \approx (0.341, 0.559)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

marks, subscripts, and units are mixed up

result written as: (0.341, 0.559); repeated-sampling interpretation (please interpret)